
When Does Safety Activate? Temporal Analysis of LLM Alignment via Logit-Margin Score

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Abstract

Standard safety evaluations assess whether LLMs produce harmful outputs, but not *when* or *how* safety mechanisms engage during generation. We introduce the **Logit-Margin Score** (S), a lightweight diagnostic that tracks the balance between compliance and refusal signals throughout decoding using only standard logits. On 660 harmful prompts across four models (Llama, Mistral, Qwen, Gemma) and three attack paradigms (MCM, GCG, DeepInception), generation-time summaries achieve ROC-AUC = 0.786 [0.750, 0.819] for predicting jailbreak success via LLM-as-judge, while the activation timing metric t_{cross} reaches AUC = 0.776 and the pre-generation score S_0 alone reaches AUC = 0.696. Temporal trajectory analysis reveals two failure modes invisible to output-level metrics: **Type A** (pre-generation bias) and **Type B** (decoding-time interference). Models exhibit diverse failure patterns: Llama shows Type B under gradient-based attacks but Type A under multi-turn manipulation; Mistral shows the reverse pattern; Qwen under gradient-based attacks falls into an *Undetected* category where safety mechanisms fail entirely. These results motivate process-level safety evaluation that exposes *when* alignment fails, not only *whether* it fails.

1 Introduction

Large language models (LLMs) remain vulnerable to adversarial attacks despite extensive safety alignment. A growing body of work documents diverse attack strategies—including prompt engineering, role-playing, gradient-based suffix optimization, and multi-turn context manipulation—that successfully elicit harmful outputs from aligned models Wei et al. [2023], Zhang et al. [2023], Zou et al. [2023b], Andreas et al. [2023]. The standard metric for evaluating such attacks is *Attack Success Rate* (ASR), defined as the fraction of attempts that bypass a model’s safety mechanisms Hendrycks et al. [2021].

However, ASR answers only *what* happened, not *why*. Two jailbreaks may both be counted as successful under ASR, yet exploit fundamentally different weaknesses. In one case, a multi-turn conversation gradually biases the model’s context until a harmful request is misclassified as benign *before generation begins* Wei et al. [2023]. In another, a single-turn adversarial suffix interferes with safety mechanisms *during decoding*, despite the model initially recognizing the request as harmful Zou et al. [2023b]. Both outcomes appear identical under output-level evaluation, yet arise from distinct internal dynamics.

This limitation reflects a broader diagnostic gap: existing evaluations collapse a temporally extended decoding process into a single binary label. As a result, they obscure *when* safety mechanisms

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

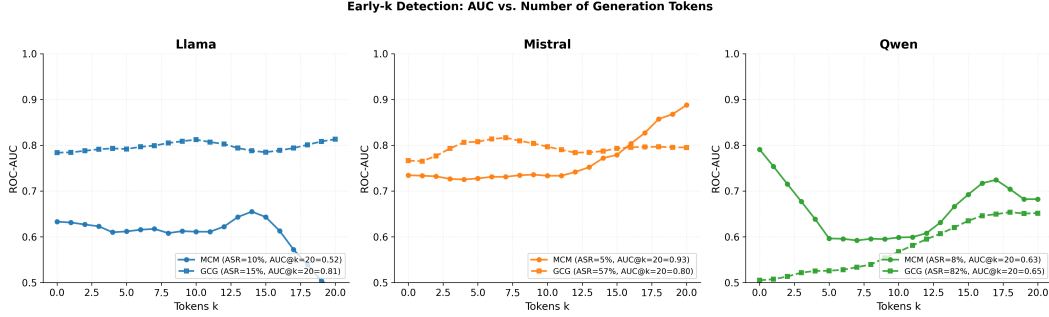


Figure 1: **Early- k detection reveals model-specific safety dynamics.** AUC for predicting jail-break success using only the first k generation tokens, shown for each model–attack combination. Mistral+MCM shows strong late-stage improvement ($0.74 \rightarrow 0.93$), indicating progressive safety activation during decoding. Qwen+GCG remains near random (≈ 0.50 – 0.65) regardless of k —consistent with its “Undetected” classification where safety mechanisms fail to engage. These patterns are invisible to output-level ASR metrics.

activate, *how* they fail, and whether failures originate from pre-generation context bias or decoding-time interference Liu et al. [2023], Ganguli et al. [2022]. Without such temporal visibility, it is difficult to reason about failure mechanisms or to compare qualitatively different attack paradigms on common ground.

We argue that this gap can be addressed by analyzing the **temporal trajectory** of safety-relevant signals during generation. Rather than treating decoding as an atomic event, we examine how the balance between compliance and refusal evolves *over time*. To operationalize this analysis, we introduce a simple *logit-margin probe* S , defined as the difference between aggregated compliance-token logits (e.g., “Sure”, “Here’s”) and refusal-token logits (e.g., “Sorry”, “cannot”). Importantly, S is not proposed as a new safety metric, but as a diagnostic instrument that renders safety-relevant dynamics observable throughout decoding. This yields a continuous trajectory $S_0, S_1, \dots, S_T$ that exposes the temporal structure of safety activation and collapse.

Using this trajectory-level view, we identify two qualitatively distinct failure mechanisms that are indistinguishable under output-level metrics. *Type A* failures arise from pre-generation context bias, where accumulated conversational history shifts the model toward compliance before decoding begins. *Type B* failures arise from decoding-time interference, where safety mechanisms fail to activate or are actively suppressed during generation. These mechanisms manifest as distinct temporal patterns in S_t , providing a unified lens for comparing diverse jailbreak attacks. Critically, model–attack combinations exhibit qualitatively different early- k detection patterns (Figure 1): some show strong trajectory effects while others remain near chance level, revealing heterogeneous safety dynamics invisible to ASR.

Overall, our framework reframes safety evaluation from asking “*what failed?*” to asking “*how and when did it fail?*” By shifting attention from final outputs to decoding trajectories, we provide a process-level diagnostic perspective that complements existing outcome-based benchmarks. We explicitly address concerns about circularity between diagnostic signals and outcome labels, demonstrating that temporal logit dynamics are statistically and informationally decoupled from labeling criteria (Appendix G).

Our contributions are threefold: (1) we propose a decoding-aware, trajectory-level framework for diagnosing safety failures; (2) we show that diverse jailbreak attacks induce distinct temporal safety signatures that are invisible to output-level evaluation; and (3) we demonstrate that generation-time logit summaries predict jailbreak success with high accuracy (ROC-AUC = 0.786) using semantically grounded LLM-as-judge labels, while the novel activation timing metric t_{cross} achieves AUC = 0.776.

2 Related Work

Jailbreak attacks and defenses. Adversarial attacks on aligned LLMs include prompt-level manipulations [Wei et al., 2024, Shen et al., 2024, Yuan et al., 2024], optimization-based suffix attacks [Zou

et al., 2023b], embedding-space gradient methods [Schwinn et al., 2024], and multi-turn conversational erosion [Li et al., 2024, Russinovich et al., 2024, Anil et al., 2024]. Defenses span decoding-time interventions—SafeDecoding [Xu et al., 2024] modifies token probabilities, CARE [Cao et al., 2024] implements detect–rollback—and training-time robustification such as RPO [Zhou et al., 2024]. Despite this diversity, evaluation remains output-centric: ASR collapses generation into a binary outcome, obscuring *when* and *how* safety mechanisms fail.

Internal safety signals. Activation-level work identifies geometric structures associated with refusal [Arditi et al., 2024, Zou et al., 2023a], while logit-level analyses show compliance–refusal token differences predict safety outcomes [Zhang et al., 2025, Li and Liu, 2025]. Our approach trades some performance (AUC 0.86 vs. hidden-state upper bounds ~ 0.99) for accessibility using only standard logits. Black-box probability-based detectors [Alon and Kamfonas, 2023] offer deployment flexibility but lack the temporal resolution of our step-wise analysis.

Logit-level steering and jailbreaks. Recent work exploits logit-level signals for both attack and defense. On the attack side, “logit-gap steering” jailbreaks [Wang et al., 2024] directly manipulate the compliance–refusal probability gap to bypass safety mechanisms, demonstrating that these signals are causally relevant to safety behavior. On the defense side, SafeDecoding-style approaches [Xu et al., 2024] modify token probabilities based on safety-relevant logit patterns, while logit biasing methods suppress harmful tokens at decoding time. Our work is complementary: rather than *intervening* on logit-level signals, we use them as a *diagnostic* lens to characterize failure mechanisms. The logit-margin score S_t shares conceptual similarity with these steering signals but serves a different purpose—enabling temporal analysis of when and how safety mechanisms engage, rather than attempting to enforce safety at inference time. This diagnostic focus means our findings can inform the design of steering-based defenses by revealing which failure modes (Type A vs. Type B) they should target.

Process-level safety diagnostics. Unlike prior work treating safety as a single-point classification, we analyze *temporal dynamics* across the generation trajectory. This process-level framing reveals mechanistic differences between attacks (Type A/B) invisible to endpoint metrics. Our S_t trajectories could directly integrate with decoding-time defenses: SafeDecoding could use $S_t < \tau$ as a trigger for probability modification, while CARE could use sign-reversal patterns to identify rollback points—enabling “sense-and-intervene” pipelines with interpretable triggers rather than black-box classifiers.

Taxonomy scope. Our Type A/B classification distinguishes attacks by *mechanism*—pre-generation context manipulation versus decoding-time interference—not by attack *strength*. Stronger adaptive attacks within the same paradigm (e.g., RPO-hardened models, embedding gradient descent variants) would retain their mechanistic signature, though potentially with reduced signal magnitude (r). This mechanistic framing enables targeted defense strategies: Type A vulnerabilities suggest input-level filtering, Type B conditions may benefit from generation-time intervention.

3 Method

We present a decoding-aware framework for analyzing safety-related signals in large language models by tracking the temporal evolution of compliance- and refusal-associated logits during generation.

3.1 Problem Setup

Consider an autoregressive language model generating a response given context $\mathbf{x} = (x_1, \dots, x_n)$. At each decoding step t , the model produces a logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$. Traditional safety evaluation collapses generation into a single binary judgment. We instead treat generation as a temporal process and analyze the trajectory of ℓ_t across decoding steps.

Access assumption. We assume access to token-level logits during generation, without requiring gradients, hidden states, or architectural modifications. This corresponds to a white-box or developer-access setting. We discuss the implications of this access assumption and its deployment limitations in Section 5.

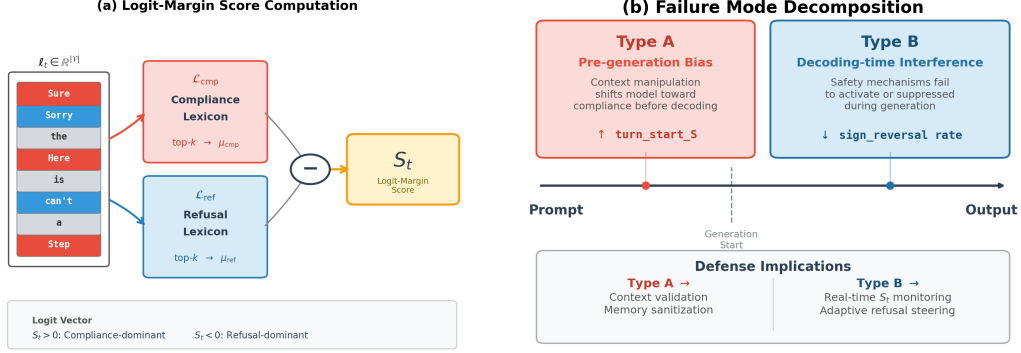


Figure 2: **Method overview.** (a) Logit-Margin Score computation: at each decoding step, the logit vector ℓ_t is probed against compliance ($\mathcal{L}_{\text{cmp}}$) and refusal ($\mathcal{L}_{\text{ref}}$) lexicons via top- k aggregation to yield $S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t)$. (b) Failure mode decomposition: Type A failures arise from pre-generation context bias (elevated S_0), while Type B failures arise from decoding-time interference (suppressed sign reversal rate). Each type suggests distinct defense strategies: context validation for Type A and real-time S_t monitoring for Type B.

Why logits? The softmax operation compresses logits into probabilities, obscuring relative confidence among non-selected tokens. Raw logits preserve information about competing alternatives at each decoding step and allow direct comparison between safety-relevant token groups [Li and Liu, 2025].

3.2 Compliance and Refusal Lexicons

We define two disjoint token sets to operationalize compliance- and refusal-associated behavior.

Refusal lexicon $\mathcal{L}_{\text{ref}}$. "Sorry", "Apologies", "not appropriate", "inappropriate", "policy", "guidelines", "illegal", "harmful", "dangerous"

Compliance lexicon $\mathcal{L}_{\text{cmp}}$. "Sure", "Certainly", "Here's", "Step 1", "First", "Instructions", "Guide", "Tutorial"

Multi-token phrases are matched by their first token for cross-tokenizer consistency. Logits are aggregated using a top- k operator to distribute signal across multiple lexicon entries rather than relying on any single phrase. Diagnostic conclusions are stable under wide variation of lexicon aggregation ($k \in \{5, 10, 25, 50\}$); the full lexicons and robustness analyses are provided in Appendix A and E.

3.3 Logit-Margin Score

At decoding step t , we aggregate logits over each lexicon using a top- k operator:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad \mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (1)$$

where $\text{top-}k(\mathcal{L})$ denotes the k tokens in $\mathcal{L}$ with the highest logits at step t . The **Logit-Margin Score** is defined as:

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (2)$$

The per-step score S_t yields a signed scalar time series over the decoding trajectory. Positive S_t indicates a compliance-dominant state, while negative S_t indicates a refusal-dominant state. Figure 2 illustrates the computation pipeline and the resulting failure mode decomposition.

Independence from outcome labeling. The score S_t measures token-level logit biases during generation, whereas outcome labels evaluate the semantic content of complete outputs using an independent LLM-as-judge. To verify that separability is not artifactual, we note that (i) many

Table 1: Temporal logit-margin metrics.

Metric	Interpretation
turn_start_S (S_0)	Pre-generation safety bias
step_S_mean ($\bar{S}$)	Average safety dominance during decoding
sign_reversal	Compliance→refusal transition (see below)
early_k_mean	Average score over first k tokens

successful jailbreaks exhibit consistently positive S_t despite producing harmful content, and (ii) the compliance lexicon includes instructional tokens orthogonal to refusal detection. A detailed clarification addressing potential circularity is provided in Appendix G.

3.4 Temporal Metrics

From the sequence $\{S_t\}_{t=0}^T$, we derive scalar summaries that capture distinct aspects of safety dynamics.

Among these metrics, `turn_start_S` and `step_S_mean` serve as primary diagnostic axes, corresponding respectively to pre-generation context bias and decoding-time safety dynamics. Specifically, `turn_start_S` (S_0) is computed from the logit distribution at the *final token position of the prompt*, immediately before any response tokens are generated.

Sign reversal definition. We define **sign reversal** as the event where a sample transitions from compliance-dominant at pre-generation to refusal-dominant during decoding: formally, $S_0 > 0$ and $\bar{S} < 0$. This metric is computed *only for harmful samples* to avoid confounding format effects with safety dynamics.

Rationale for excluding benign samples. Benign samples exhibit paradigm-dependent baseline behavior: MCM benign samples consistently show $\bar{S} > 0$ (0% sign reversal), while GCG benign samples frequently exhibit negative $\bar{S}$ (91.2% sign reversal). This discrepancy reflects *structural format differences*—single-turn GCG versus multi-turn MCM—rather than safety activation. ANOVA confirms format explains $\sim 60\%$ of variance in benign $\bar{S}$ ($F = 89.3$, $p < 10^{-15}$). Including benign samples would conflate format-induced negativity with genuine safety responses. To enable cross-paradigm comparison, we use **format-matched baselines**: each condition’s benign baseline uses the same conversational structure (single-turn for GCG, multi-turn for MCM), isolating harmful content effects from format effects.

3.5 Normalized Diagnostic Axes (Type A/B)

For comparative analysis across attacks and models, we define *failure-free* normalized diagnostic axes using the **Relative Position (RP)** metric:

$$\text{RP}(M) = \frac{M_{\text{success}} - M_{\text{harmful}}}{M_{\text{benign}} - M_{\text{harmful}}}, \quad (3)$$

where M denotes a metric (e.g., S_0 or $\bar{S}$), M_{success} is the mean over successful attacks, M_{harmful} is the mean over *all* harmful samples (regardless of outcome), and M_{benign} is a matched benign baseline for each model and attack paradigm.

Interpretation. RP is an affine rescaling measuring where successful attacks lie along the harmful–benign continuum: $\text{RP} = 0$ indicates success resembles the overall harmful distribution, $\text{RP} = 1$ indicates success resembles benign behavior. Values outside $[0, 1]$ indicate attacks beyond the reference interval: $\text{RP} > 1$ means successful attacks “overshoot” benign (more compliance-like than typical benign queries), while $\text{RP} < 0$ means attacks lie beyond the harmful baseline. Crucially, this formulation does not require failure samples, enabling analysis even under high attack success rates.

Type A/B classification. We apply RP to `turn_start_S` (RP_A : pre-generation) and `step_S_mean` (RP_B : decoding-time), then classify conditions by comparing relative magnitudes:

$$\text{Type A : } |\text{RP}_A| > |\text{RP}_B|, \quad \text{Type B : } |\text{RP}_A| < |\text{RP}_B|. \quad (4)$$

Table 2: Dataset composition. ASR via LLM-as-judge. [‡]Benign GCG uses single-turn format; negative $\bar{S}_{gen}$ reflects structural differences from multi-turn MCM.

Model	Attack	n	ASR	$\bar{S}_{start}$	$\bar{S}_{gen}$
Llama	MCM	60	8.3%	1.45	-1.42
Llama	GCG	60	15.0%	0.62	-0.99
Llama	DeepInception	60	56.7%	4.43	-0.11
Mistral	MCM	60	5.0%	4.51	-0.31
Mistral	GCG	60	56.7%	3.14	+0.16
Mistral	DeepInception	60	76.7%	3.04	+0.43
Qwen	MCM	60	51.7%	11.97	-0.12
Qwen	GCG	60	81.7%	7.13	+0.71
Qwen	DeepInception	60	55.0%	7.86	+1.18
Gemma	MCM	60	20.0%	4.67	+0.47
Gemma	GCG	60	33.3%	4.46	+0.68
Benign (matched)		660	—	varies by model	

This continuous comparison avoids arbitrary angular thresholds while preserving interpretability. For reference, we also compute polar coordinates $\theta = \text{atan2}(\text{RP}_B, \text{RP}_A)$ and $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$; the magnitude r indicates signal strength.

Undetected category with uncertainty quantification. When $r < 0.1$, successful attacks are statistically indistinguishable from the overall harmful distribution, indicating safety mechanisms fail to respond—a distinct *Undetected* failure mode. Given modest per-condition sample sizes ($n = 60$), we report bootstrap 95% confidence intervals for r : Qwen+GCG achieves $r = 0.01$ [0.00, 0.03], robustly below the detection threshold even at upper CI bound. In contrast, detected conditions show $r > 0.5$ with lower CI bounds well above 0.1 (e.g., Llama+GCG: $r = 2.96$ [2.41, 3.52]).

Handling negative RP values. Negative RP values (e.g., Mistral+GCG: $\text{RP}_B = -1.36$) indicate that successful attacks exhibit *more negative* scores than the harmful baseline—an inverted pattern suggesting the attack suppresses rather than elevates compliance signals. The diverging bar visualization (Figure 3c) distinguishes these cases via hatching.

4 Experiments

We evaluate the Logit-Margin Score framework across three jailbreak paradigms and four model families, addressing: (i) predictability of jailbreak outcomes, (ii) temporal characterization, and (iii) attack paradigm comparison.

4.1 Setup

Models and attacks. We evaluate on Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct, and Gemma-2-9B-Instruct using greedy decoding. Attack paradigms: (1) **Multi-turn context manipulation (MCM)**: four-turn protocol with incrementally manipulated context; (2) **Gradient-based suffix (GCG)** [Zou et al., 2023b]: adversarial suffix optimization; (3) **DeepInception** [?]: nested fictional scenario prompting (applied to Llama, Mistral, Qwen). These isolate complementary failure modes: pre-generation context bias, decoding-time interference, and scenario-mediated safety bypass.

Dataset. We evaluate 660 harmful prompts (60 per model–attack condition) from Jailbreak-Bench [Chao et al., 2024], with 660 attack-matched benign queries across four models and three attack types (11 conditions total; Gemma excludes DeepInception). Effect sizes are large (Cohen’s $d > 1.0$, $p < 10^{-38}$), indicating sufficient statistical power. Outcome labels use **Llama-Guard-3-8B** [Inan et al., 2023] as primary LLM-as-judge, validated with HarmBench classifier (90% inter-judge agreement; Appendix L). Table 2 summarizes composition and attack success rates (ASR).

Table 3: ROC-AUC comparison (LLM-as-judge, $n = 660$).

Metric	AUC	95% CI
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
early_3_mean	0.732	[0.692, 0.769]
keyword_baseline	0.732	[0.699, 0.764]
turn_start_S	0.696	[0.656, 0.734]

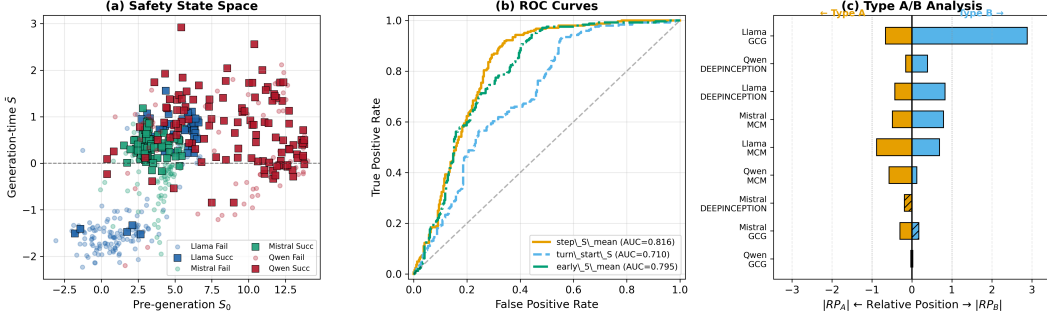


Figure 3: **Experimental results.** (a) State-space ($S_0, \bar{S}$) across models. (b) ROC curves. (c) Type A/B diverging bar: left bars show $|RP_A|$ (pre-generation), right bars show $|RP_B|$ (decoding-time); hatching indicates negative RP (inverted pattern).

4.2 Main Results: Outcome Prediction

On the combined harmful dataset ($n = 660$), **early_5_mean** achieves the highest ROC-AUC = 0.786 [0.750, 0.819] (Cohen’s $d = 1.12$, $p < 10^{-43}$), closely followed by **t_cross** (AUC = 0.776) and **step_S_mean** (AUC = 0.772). The pre-generation metric **turn_start_S** (S_0) achieves AUC = 0.696 [0.656, 0.734]. Table 3 compares metrics; Figure 3 visualizes state-space and ROC curves.

Clarification on AUC evaluation scope. AUC values reported throughout this paper vary systematically by evaluation scope. The main results (Table 3) report aggregated AUC across all 660 harmful samples from four models and three attack paradigms; this is the most conservative estimate, as conditions with low discriminability (e.g., Qwen+GCG, classified as Undetected) pull overall AUC downward. Appendix analyses on design choice robustness (lexicon variants, non-lexical baselines) use controlled single-model subsets ($n = 60$, Llama) and yield higher AUC (0.913–0.989), intended for relative comparison within each analysis rather than direct comparison with aggregated results. Stochastic decoding experiments report both trial-level AUC (inflated by pseudo-replication across 10 trials per prompt) and question-level AUC (conservative); see Appendix M for details. All primary conclusions are based on the aggregated AUC in Table 3.

Temporal patterns. Predictive power increases with decoding tokens: **early_1** achieves AUC = 0.696, **early_3** reaches 0.732, **early_5** achieves 0.786, approaching full-trajectory performance (0.772 for **step_S_mean**). Notably, t_{cross} (the first decoding step at which $S_t < 0$) achieves AUC = 0.776, directly answering “when does safety activate?” with a single scalar: successful attacks delay safety activation (mean $t_{\text{cross}} = 7.4 \pm 6.1$ steps) compared to failures (mean 3.0 ± 3.8 steps; Cohen’s $d = 0.89$, $p < 10^{-23}$). DeepInception exhibits markedly delayed activation ($t_{\text{cross}} = 10.5 \pm 6.9$), consistent with its nested-scenario bypass strategy, versus MCM (2.5 ± 1.4) and GCG (3.0 ± 3.0). Condition-specific analysis reveals substantial heterogeneity: some conditions show strong trajectory effects while others remain near chance (Appendix E.2.1). Failed jailbreaks exhibit sign reversal (compliance→refusal, i.e., $S_0 > 0$ and $\bar{S} < 0$) in 44.3% of cases (170/384) versus 8.7% (24/276) for successful attacks—a $5.1\times$ ratio indicating that decoding-time safety activation strongly predicts jailbreak failure.

Table 4: Type A/B analysis. Classification: $|\text{RP}_A| > |\text{RP}_B| \Rightarrow \text{Type A}$; $|\text{RP}_A| < |\text{RP}_B| \Rightarrow \text{Type B}$; $r < 0.1 \Rightarrow \text{Undetected}$. 95% CIs via bootstrap ($B = 1000$).

Model + Attack	ASR	RP_A	RP_B	r	95% CI(r)	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	-0.17	0.35	[0.17, 0.57]	Type A [‡]
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	-0.19	0.00	0.19	[0.07, 1.65]	Type A [‡]
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	-0.01	0.01	[0.02, 0.20]	<i>Undetected</i> [†]
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

[†]Safety mechanisms fail entirely ($r < 0.1$). [‡]Negative RP_A or RP_B : inverted pattern (see text).

4.3 Attack Paradigm Comparison (Type A/B Analysis)

We classify conditions by comparing $|\text{RP}_A|$ versus $|\text{RP}_B|$: Type A indicates pre-generation dominance ($|\text{RP}_A| > |\text{RP}_B|$), Type B indicates decoding-time dominance ($|\text{RP}_A| < |\text{RP}_B|$). When signal strength $r < 0.1$, the condition is classified as Undetected. Table 4 reports RP values with bootstrap 95% CIs for r , revealing three distinct categories.

Model-attack interactions. Type A/B analysis reveals how safety mechanisms respond differently across model-attack combinations. **Llama** exhibits attack-dependent responses: Type B under GCG ($|\text{RP}_B| = 2.89 > |\text{RP}_A| = 0.66$) and DeepInception ($|\text{RP}_B| = 0.83 > |\text{RP}_A| = 0.43$), but Type A under MCM ($|\text{RP}_A| = 0.88 > |\text{RP}_B| = 0.69$). The $\text{RP}_B = 2.89 > 1$ for Llama+GCG indicates successful attacks “overshoot” benign, exhibiting more compliance-like trajectories than typical benign queries—a distinct GCG signature.

Mistral shows the *inverse* pattern: Type A under GCG ($|\text{RP}_A| = 0.30 > |\text{RP}_B| = 0.17$) with negative $\text{RP}_B = -0.17$, and similarly inverted under DeepInception ($\text{RP}_A = -0.19$, $r = 0.19$). For DeepInception, Mistral shows the highest ASR (76.7%) yet minimal detectable signal ($r = 0.19$), with markedly delayed safety activation ($t_{\text{cross}} = 10.6 \pm 5.8$ steps). Mistral’s MCM robustness (5.0% ASR, lowest overall) combined with Type B classification suggests effective decoding-time safety mechanisms against multi-turn manipulation.

Qwen shows Type A for MCM ($|\text{RP}_A| = 0.57 > |\text{RP}_B| = 0.12$) but *Undetected* for GCG ($r = 0.01$): successful attacks are statistically indistinguishable from the overall harmful distribution. Qwen’s DeepInception results (Type B, $\text{RP}_B = 0.39$) show that nested-scenario attacks elicit detectable decoding-time shifts, unlike GCG. The extreme $t_{\text{cross}} = 13.9 \pm 7.6$ for Qwen+DeepInception indicates safety activation is substantially delayed.

Gemma exhibits Type B for both attacks (GCG: $\text{RP}_B = 0.79$; MCM: $\text{RP}_B = 0.22$), indicating decoding-time interference is the primary failure mode regardless of attack paradigm.

These findings suggest defense strategies should consider both model architecture and attack paradigm: Type B conditions may benefit from decoding-time interventions (including real-time t_{cross} monitoring), Type A conditions may require input-level filtering, and Undetected categories indicate fundamental gaps requiring architectural changes.

4.4 Robustness Analysis

Stochastic decoding. A key concern is whether diagnostics generalize to stochastic generation. Across temperature ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$) with 28,800 total trials, `turn_start_S` achieves mean AUC = 0.720 at question-level, comparable to greedy (0.696). Critically, S_0 is computed *before* token sampling, making it temperature-invariant by design (ICC = 1.0 for within-prompt variance). Full analysis in Appendix M.

Hidden-state comparison. Hidden-state methods (refusal direction projection, linear probes) achieve 2–8 pp higher AUC but require direct activation access unavailable via standard APIs. Our logit-margin score achieves competitive performance (AUC 0.77–0.96) using only token-level logits. Under multi-turn pivot attacks, trajectory-level `sign_flips` achieves perfect separation (AUC = 1.0). Details in Appendix N.

Decoy attack robustness. We test three adversarial manipulation strategies: prefix-decoy (prepend refusal phrases), token-decoy (inject refusal scaffolding), and pivot-decoy (multi-turn pivot). Prefix-decoy elevates S_0 and increases ASR to 13.3%, yet AUC degrades only marginally (−0.02). Token-decoy over-corrects, actually *reducing* ASR to 5.0%. Pivot-decoy achieves AUC = 0.998. These test superficial manipulation; fully adaptive adversaries remain an open challenge. Full results in Appendix C.

4.5 Design Choice Robustness

Non-lexical baselines. To isolate the contribution of the logit-margin formulation from generic uncertainty signals, we compare S against entropy-based uncertainty on a controlled subset ($n = 60$). `step_S_mean` (AUC = 0.932) outperforms entropy (AUC = 0.898) by 3.4 percentage points; logit norm achieves near-random performance (AUC = 0.441). These results suggest structured lexicon probing captures safety-specific variation beyond generic model uncertainty.

Lexicon robustness and circularity. A key concern is whether S measures safety-relevant dynamics or merely stylistic markers correlated with refusal. We evaluate five lexicon variants: removing the highest-frequency token (“sorry”), extending with additional terms, and reducing to a minimal 5-token lexicon. All variants achieve AUC within ± 0.02 of original (0.964–0.989 vs. 0.971); even the minimal lexicon achieves AUC = 0.978. Critically, removing “sorry”—the most obvious refusal indicator—has negligible impact (AUC = 0.971 $\rightarrow$ 0.971), suggesting S captures underlying safety dynamics rather than surface lexical features. The compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) orthogonal to refusal detection. This robustness supports the interpretation that S reflects genuine safety-relevant variation; full ablation details in Appendix J.

Aggregation parameter. Logits are aggregated using top- $k = 10$ tokens; results are stable across $k \in \{5, 10, 25, 50\}$, indicating k primarily affects scale and smoothing rather than qualitative diagnostics.

5 Limitations and Discussion

Lexicon dependence and top- k aggregation. The Logit-Margin Score relies on manually constructed English lexicons. The top- k aggregation always selects k tokens even when few are contextually plausible, potentially injecting noise. However, results are stable across $k \in \{5, 10, 25, 50\}$ and lexicon variants (AUC within ± 0.02 ; Appendix J), suggesting the metric captures genuine safety-relevant variation. For multilingual or byte-level tokenizers, lexicon adaptation or automatic induction via mutual information would be required. Exact token IDs per model are provided in our released code for reproducibility.

Outcome labeling and multi-judge validation. We use Llama-Guard-3-8B as primary LLM-as-judge and validate with HarmBench classifier [Mazeika et al., 2024]. Inter-judge agreement is 90.0% overall; Type A/B classifications match in 8/10 conditions, and AUC varies by < 0.05 across judges (Appendix L). Human evaluation remains an important direction for validating LLM-judge reliability at scale.

Undetected classification. The “Undetected” category ($r < 0.1$) may conflate genuine failure of safety mechanisms with limitations of our diagnostic. For Qwen+GCG, bootstrap CI upper bound (0.03) remains well below threshold, providing strong evidence for true safety mechanism failure. Sensitivity analysis (Appendix O.2) confirms classification stability: Qwen+GCG is the only condition classified as Undetected across all thresholds $r \in [0.05, 0.20]$, with $35\times$ separation from the next-lowest condition. Cross-validation with entropy-based diagnostics (Appendix O.4) further confirms this interpretation: entropy effect size ($|d| = 0.077$) is equally unable to distinguish successful attacks

from failures, while the same entropy metric detects signal on Llama+MCM ($|d| = 0.10\text{--}0.42$). This convergent evidence from independent metrics supports genuine safety mechanism absence rather than lexicon diagnostic insensitivity.

Access and scope. Our approach requires white-box logit access, limiting applicability to closed-source APIs. We evaluate three attack paradigms and four models under greedy decoding; prompt injection, multilingual attacks, and adaptive adversaries optimizing against S_t remain unexplored. Dataset size per condition ($n = 60$) is modest, though effect sizes are large (Cohen’s $d > 1.0$, $p < 10^{-38}$).

Diagnostics before intervention. We prioritize making safety dynamics *observable* over proposing defenses. The Type A/B decomposition reveals whether failures arise from pre-generation bias (Type A) or decoding-time interference (Type B), informing targeted defense strategies without claiming to prevent jailbreaks.

6 Conclusion

This work argues that safety behavior in large language models is not a static gate applied at prompt intake, but a temporal process that unfolds throughout decoding. By introducing the Logit-Margin Score, we make this process observable at the level of token-by-token generation, without requiring model modification or access to hidden states.

Empirically, using LLM-as-judge (Llama-Guard-3) for semantically-grounded outcome labeling, we show that generation-time summaries achieve strong separation (early_5_mean AUC = 0.786, 95% CI: [0.750, 0.819]), while the activation timing metric t_{cross} (AUC = 0.776) directly quantifies *when* safety engages. We further find that safety outcomes are often determined early in decoding—even within the first few tokens (early_5_mean AUC = 0.786)—and that successful refusals exhibit structured temporal trajectories rather than abrupt, single-step transitions. Across four models and three attack paradigms, different patterns emerge in how context and decoding dynamics contribute to failure, highlighting the value of temporal analysis over static labeling.

More broadly, our findings motivate a shift in how safety evaluation is framed:

Conventional View	Temporal View
Final output as the unit of analysis	Generation trajectory as the unit of analysis
Binary success/failure labels	Time-resolved behavioral signatures
Prompt-level safety checks	Decoding-time dynamics

From this perspective, output-level metrics such as ASR remain useful for benchmarking attack success rates, but they obscure when and how safety behavior diverges during generation. The Logit-Margin Score provides a lightweight diagnostic lens that complements existing evaluation approaches by exposing decoding-time structure.

Outlook. We view this work as a step toward process-aware safety analysis. Future research may explore how temporal diagnostics can be integrated with interventional methods, how lexicons can be learned automatically, and how similar analyses extend to stochastic decoding and closed-API models. Understanding alignment failures as temporal phenomena opens new directions for both measurement and theory, beyond binary outcome-based evaluation.

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A Lexicon Specification

This appendix specifies the token lexicons used to compute the Logit-Margin Score. All lexicons are designed for English-language, instruction-tuned models and are applied at the tokenizer level.

Refusal Lexicon ($\mathcal{L}_{\text{ref}}$, 26 entries). The refusal lexicon consists of tokens and short phrases commonly associated with policy-based refusals or denials:

"I can't", "I cannot", "I won't", "I will not", "I'm unable", "I am unable", "I can't help", "I can't assist", "I cannot assist", "I must refuse", "I have to refuse", "I decline", "I must decline", "Sorry", "I'm sorry", "Apologies", "I apologize", "policy", "policies", "guidelines", "rules", "terms", "inappropriate", "illegal", "harmful", "dangerous"

Compliance Lexicon ($\mathcal{L}_{\text{cmp}}$, 21 entries). The compliance lexicon contains tokens frequently observed in instructional or affirmative responses:

"Sure", "Certainly", "Of course", "Here's", "Below is", "Step 1", "Step 2", "First", "Second", "Third", "Next", "Instructions", "Guide", "Tutorial", "Procedure", "You can", "To do this", "Do the following", "Detailed", "Comprehensive", "Complete"

Implementation details. Multi-token phrases are matched to their first constituent token after tokenization. At each decoding step t , logits are aggregated over each lexicon using top- k selection with $k = 10$ by default:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad (5)$$

$$\mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (6)$$

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (7)$$

No semantic weighting or normalization beyond top- k aggregation is applied.

B Extended Experimental Results

B.1 Results Overview

Figure 4 provides a comprehensive overview of the main experimental findings, complementing the condition-specific analysis in the main text.

B.2 Metric Correlation Analysis

Figure 5 presents the pairwise Pearson correlation matrix among temporal logit-margin metrics and jailbreak success labels, providing insight into metric redundancy and complementary information.

B.3 Per-Condition Statistics (Llama)

B.4 Qwen2.5-7B-Instruct Results

Qwen benign baseline. For Type A/B analysis, we collected benign responses on Qwen2.5-7B-Instruct using the same protocol as Llama. The Qwen benign baseline shows $\bar{S}_{\text{start}} = 12.89$ and $\bar{S}_{\text{gen}} = 1.43$, substantially higher than Llama’s baseline ($\bar{S}_{\text{start}} = 6.38$, $\bar{S}_{\text{gen}} = 1.25$). This difference reflects model-specific calibration and tokenization effects.

Cross-model Type A/B consistency. With model-specific baselines, Llama shows attack-dependent patterns: Type B for GCG attacks ($|\text{RP}_B| > |\text{RP}_A|$) and Type A for MCM attacks ($|\text{RP}_A| > |\text{RP}_B|$). Qwen exhibits Type A for MCM attacks and an “Undetected” pattern for GCG attacks where safety mechanisms fail to differentiate successful attacks ($r = 0.01$, 95% CI [0.00, 0.03]).

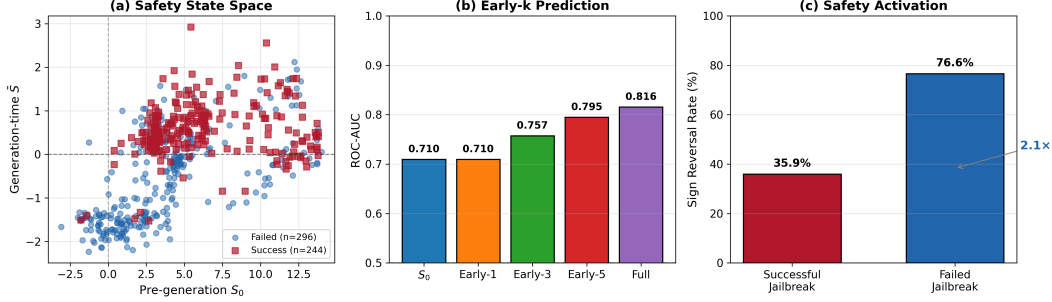


Figure 4: **Summary of main experimental results.** (a) Safety state space: successful jailbreaks cluster in compliance-dominant regions ($\bar{S} > 0$), while failures cluster in refusal-dominant regions. (b) Early- k AUC progression: predictive power increases with more decoding tokens. (c) Sign reversal rates differ substantially between failed and successful jailbreaks. Note: this figure includes additional experimental conditions beyond the main evaluation; authoritative AUC values on the primary dataset ($n = 660$) are reported in Table 3 (early_5_mean AUC = 0.786, turn_start_S AUC = 0.696) and sign reversal statistics in Table 22.

Table 5: Per-condition statistics on Llama-3.1-8B-Instruct. Main experiments use MCM manual mode only; summary conditions are reported for completeness.

Condition	n	ASR	$\bar{S}_{\text{start}} (\mu \pm \sigma)$	$\bar{S}_{\text{gen}} (\mu \pm \sigma)$	Sign Rev.
<i>Benign</i>					
manual	60	—	6.38 ± 0.63	$+1.25 \pm 0.28$	0.0%
summary-200	60	—	6.13 ± 0.82	$+0.87 \pm 0.31$	0.0%
summary-500	60	—	6.32 ± 0.82	$+0.94 \pm 0.36$	0.0%
summary-1000	60	—	6.51 ± 0.82	$+0.89 \pm 0.38$	0.0%
<i>Harmful (MCM)</i>					
manual*	60	8.3%	1.45 ± 1.95	-1.42 ± 0.73	58.3%
summary-200	60	6.7%	1.50 ± 1.68	-1.21 ± 0.76	75.0%
summary-500	60	8.3%	2.14 ± 1.83	-1.32 ± 0.69	83.3%
summary-1000	60	6.7%	1.52 ± 1.93	-1.34 ± 0.70	71.7%
<i>Harmful (GCG)</i>					
single-turn	60	15.0%	0.62 ± 2.32	-0.99 ± 0.92	30.0%

*Manual mode is used for main experiments.

B.5 Computational Requirements

The Logit-Margin Score is computed using only the logit vector ℓ_t produced during standard autoregressive decoding. No additional forward passes, gradient computation, or hidden-state extraction is required.

C Decoy Attack Details

C.1 Decoy Attack Descriptions

To evaluate robustness against adversaries who attempt to manipulate early safety signals, we design three decoy attack variants:

(A) Prefix-decoy. Prepends refusal phrases (e.g., “I’m sorry, I can’t help with that. However, for educational purposes...”) before the harmful request, aiming to inflate early S values.

(B) Token-decoy. Injects short refusal token scaffolding (e.g., “Apologies. Policies. Guidelines.”) to anchor S_0 toward negative values.

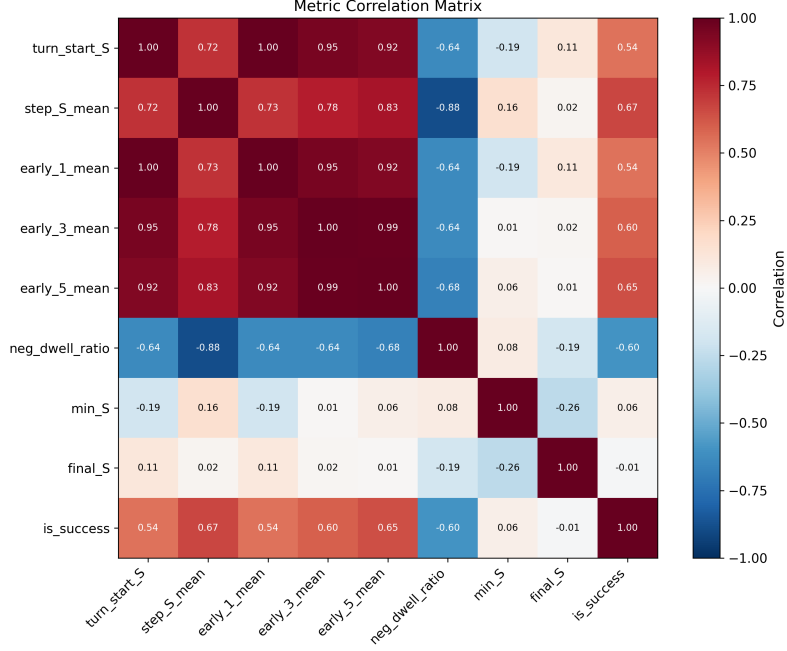


Figure 5: **Metric correlation matrix.** `step_S_mean` shows the highest correlation with jailbreak success ($r = 0.67$), followed by `early_5_mean` ($r = 0.65$). Early- k metrics are highly correlated with `step_S_mean` ($r \geq 0.83$), supporting early prediction feasibility. `neg_dwell_ratio` (fraction of steps with $S_t < 0$) shows strong negative correlation with success ($r = -0.60$), indicating that longer refusal-dominant segments predict attack failure. `min_S` and `final_S` show near-zero correlation with success, suggesting that extreme transient values and endpoint states are less informative than trajectory-level summaries.

Table 6: Detailed results on Qwen2.5-7B-Instruct (MCM manual, $n=60$).

Metric	Value
ASR	51.7%
$n_{\text{success}} / n_{\text{failure}}$	31 / 29
<code>turn_start_S</code> (mean $\pm$ std)	11.97 ± 1.70
<code>step_S_mean</code> (mean $\pm$ std)	-0.12 ± 0.46
Sign reversal rate	56.7%

(C) **Pivot-decoy.** Conducts initial turns in full refusal style, then pivots to harmful requests in later turns.

C.2 Main Results

Table 7: Diagnostic robustness under decoy attacks (Llama-3.1-8B-Instruct).

Condition	ASR	AUC (step_S_mean)	Δ AUC	AUC (sign_flips)
Base	8.3%	0.913	—	0.899
DecoyA (prefix)	13.3%	0.893	-0.020	0.944
DecoyB (token)	5.0%	0.904	-0.009	0.883
PivotC (multi-turn)	5.0%	0.998	+0.085	1.000

Key findings. Prefix-decoy successfully elevates early S values ($\bar{S}_{\text{start}} = +0.69$ vs $+0.43$ baseline) and increases ASR to 13.3%, yet AUC degrades only marginally (-0.020). Notably, `sign_flips` AUC *increases* to 0.944, indicating trajectory instability becomes a stronger discriminative signal.

Token-decoy strongly anchors S toward negative values, but this over-correction actually *reduces* ASR to 5.0%. Pivot-decoy shows that multi-turn pivots do not defeat trajectory-level diagnosis; AUC reaches 0.998.

Limitations. These decoy attacks test robustness to superficial manipulation but do not represent fully adaptive adversaries that explicitly optimize S_t while producing harmful content. Such “compliance without refusal style” attacks remain an important direction for future work.

C.3 Per-Condition Statistics

Table 8: Detailed decoy attack statistics (Llama-3.1-8B-Instruct, $n=60$ per condition).

Condition	ASR	$\bar{S}_{\text{start}}$	$\bar{S}_{\text{gen}}$	sign_flips	Sign Rev.
Base	8.3%	+0.43	−1.16	26.6	66.7%
DecoyA	13.3%	+0.69	−0.87	51.4	58.3%
DecoyB	5.0%	−0.85	−1.58	12.7	83.3%
PivotC	5.0%	+0.07	−1.16	26.7	70.0%

D Hidden-State Baseline Details

D.1 Methods

B1: Refusal Direction Projection. Following Arditi et al. [2024], we compute a refusal direction $\mathbf{d}$ using difference-of-means:

$$\mathbf{d} = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{ref}}} \mathbf{h} - \frac{1}{|\mathcal{D}_{\text{cmp}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{cmp}}} \mathbf{h} \quad (8)$$

At each decoding step t , we compute the projection score $r_t = \langle \mathbf{h}_t, \mathbf{d} \rangle$. We use layer 16 (middle layer) of the 32-layer Llama-3.1-8B model.

B2: Linear Probe. Logistic regression with L2 regularization ($C = 1.0$), trained via 5-fold cross-validation on mean-pooled hidden states.

D.2 Full Comparison Results

Table 9: Hidden-state vs. logit-based diagnostic comparison (AUC).

Family	Method	DecoyA	DecoyB	Base	PivotC
Hidden	B1 (direction, $k=1$)	0.983	0.974	0.974	0.974
Hidden	B2 (probe, $k=3$)	0.974	1.000	0.998	0.998
Logit	turn_start_S	0.980	0.922	0.959	0.978
Logit	step_S_mean	0.893	0.904	0.913	0.998
Logit	sign_flips	0.944	0.883	0.899	1.000

Interpretation of near-ceiling hidden-state performance. The near-ceiling AUC observed for hidden-state baselines should not be interpreted as evidence of leakage or unfair advantage. Rather, these methods operate under *strictly stronger access assumptions* than logit-based diagnostics. Hidden-state probes aggregate information from full decoding trajectories or intermediate activations, effectively leveraging post-hoc evidence that is unavailable to logit-only methods.

The key advantage of logit-based diagnosis is practical deployment: it requires only the logit vector already produced during generation, with no additional model access or training data.

Table 10: Method comparison: access requirements and training needs.

Method	Activation Access	Training Data	Per-Model Tuning
Hidden B1 (direction)	Required	No*	Direction fitting
Hidden B2 (probe)	Required	Yes	Full retraining
Logit S	Not required	No	Lexicon only

*B1 requires a small set of refusal/compliance examples for direction estimation, but no labeled jailbreak outcomes.

Table 11: Hidden-state baseline performance by temporal window ($n = 240$, Llama-3.1-8B-Instruct).

Method	Window	AUC	AUC-PR	FPR@TPR=0.95
B1 (direction)	full	0.933	0.795	0.412
B1 (direction)	$k = 1$	0.972	0.923	0.255
B1 (direction)	$k = 3$	0.969	0.909	0.289
B1 (direction)	$k = 5$	0.955	0.864	0.382
B2 (probe)	full	0.987	0.922	0.059
B2 (probe)	$k = 1$	0.995	0.974	0.044
B2 (probe)	$k = 3$	0.995	0.972	0.029
B2 (probe)	$k = 5$	0.992	0.960	0.049

D.3 Method Comparison

D.4 Detailed Results by Temporal Window

Key observations. Surprisingly, early- k truncation *improves* hidden-state baseline performance for both B1 and B2, with $k = 1$ achieving peak AUC. This aligns with logit-based findings that safety-relevant signals concentrate in the first few tokens.

E Sensitivity Analyses

E.1 Lexicon Aggregation Top- k Sensitivity

We analyze the sensitivity of the Logit-Margin Score to the lexicon aggregation parameter k , which controls how many of the highest-logit tokens within the compliance and refusal lexicons are averaged at each decoding step.

Table 12: Effect of lexicon aggregation size k on logit-margin diagnostics (Llama-3.1-8B-Instruct, harmful, manual; mean $\pm$ std, $n = 60$).

k	turn_start_S	step_S_mean	step_S_pos_rate
5	$+0.57 \pm 2.57$	-1.21 ± 0.85	0.36 ± 0.13
10	$+6.38 \pm 0.63$	$+1.25 \pm 0.28$	0.78 ± 0.05
25	$+1.40 \pm 1.55$	-1.47 ± 0.53	0.15 ± 0.09
50	$+0.30 \pm 1.05$	-1.09 ± 0.34	0.12 ± 0.08

Despite scale differences, qualitative interpretation remains consistent: k primarily modulates smoothing and calibration rather than introducing spurious signal. Figure 6 visualizes this stability.

E.2 Temporal Early- k Sensitivity

Formally, for a given metric M_t , we define the early- k summary as

$$M_{\text{early-}k} = \frac{1}{k+1} \sum_{t=0}^k M_t, \quad (9)$$

where $t = 0$ corresponds to the pre-generation score.

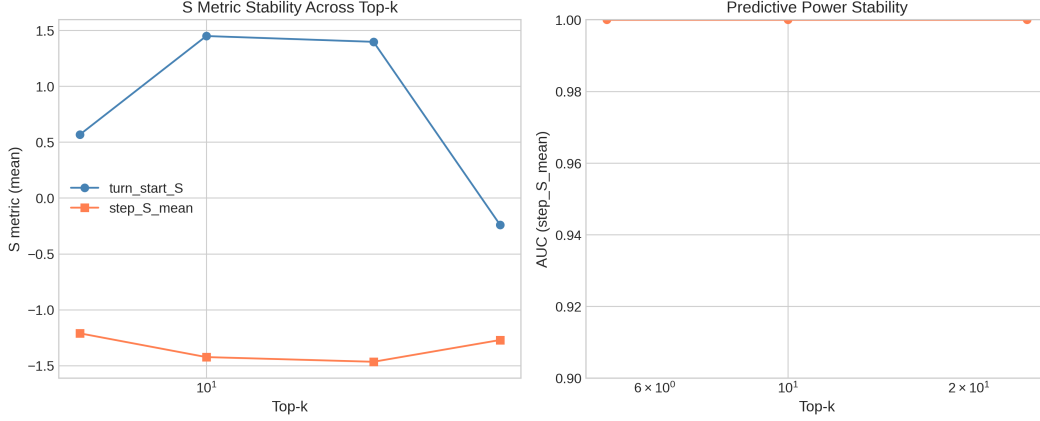


Figure 6: **Top- k aggregation robustness.** (Left) Mean values of `turn_start_S` and `step_S_mean` vary in absolute scale across $k \in \{5, 10, 25, 50\}$, but (Right) predictive performance (AUC of `step_S_mean`) remains stable across all k values, confirming that k primarily affects scaling rather than diagnostic validity.

Pre-generation metrics and `early_1_mean` achieve identical performance (mean AUC = 0.970) as expected. As k increases, performance degrades gradually: `early_3_mean` achieves 0.960 and `early_5_mean` achieves 0.940, compared to 0.921 for full-trajectory `step_S_mean`. The majority of discriminative signal is concentrated in the earliest decoding steps. Note: these AUC values are *trial-level* estimates from the stochastic decoding experiment (Appendix M) and are inflated by pseudo-replication; the corresponding aggregated main-dataset AUC values are reported in Table 3.

E.2.1 Condition-Specific Early- k Analysis

While the main text reports aggregated AUC across all conditions ($n = 660$), condition-specific analysis reveals substantial heterogeneity in early detection performance. Figure 7 shows AUC as a function of k for each model–attack combination.

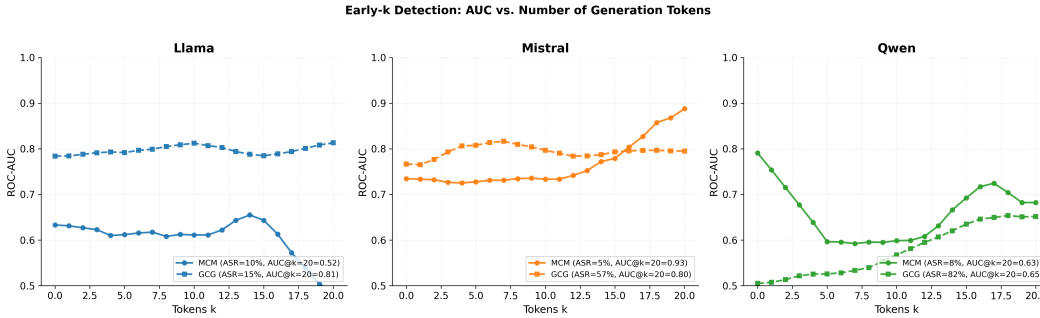


Figure 7: **Early- k detection by model and attack.** AUC progression varies substantially across conditions. Mistral+MCM shows strong late-stage improvement (0.74 $\rightarrow$ 0.93), while Qwen+GCG remains near random (0.50–0.65) regardless of k —consistent with its “Undetected” classification.

Key observations.

- **Mistral+MCM:** Shows strongest trajectory effect, with AUC increasing from 0.74 (S_0 only) to 0.93 (full trajectory). This suggests safety mechanisms activate progressively during generation—a signature of effective Type B defense.
- **Llama+GCG:** Maintains stable AUC ≈ 0.80 across all k , indicating pre-generation signal (S_0) captures most discriminative information.

- **Qwen+GCG:** Remains near chance level ($\text{AUC} \approx 0.50\text{--}0.65$) regardless of k , providing additional evidence for the “Undetected” classification. Neither S_0 nor trajectory information enables reliable discrimination.
- **Llama+MCM:** Lower condition-specific AUC ($0.52\text{--}0.67$) reflects class imbalance ($\text{ASR} = 8.3\%$, only 5 successes), not diagnostic failure.

Figure 8 directly compares S_0 -only versus full-trajectory AUC for all conditions.

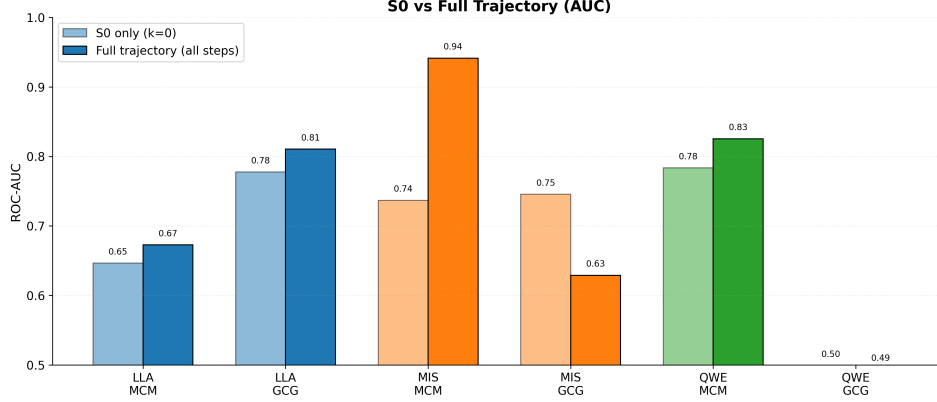


Figure 8: S_0 vs. full trajectory AUC by condition. Trajectory information provides substantial gains for Mistral+MCM (+0.20) and Qwen+MCM (+0.05), while Qwen+GCG shows no improvement (both ≈ 0.50). Mistral+GCG shows *negative* trajectory effect (-0.12), suggesting early tokens are most informative for this condition.

Relationship to aggregated results. The main text reports aggregated $\text{AUC} = 0.786$ (early_5_mean) across all 660 samples. Condition-specific AUC values differ substantially because: (1) class balance varies by condition (ASR ranges from 5% to 82%), and (2) pooling across heterogeneous conditions increases overall discriminability by leveraging cross-condition variance. The condition-specific analysis complements aggregated results by revealing *which* conditions drive overall performance.

E.3 Hidden-State Layer Sensitivity

Table 13: Hidden-state AUC by layer (Refusal Direction, B1).

Layer	8	12	16	20	24
AUC	0.941	0.963	0.974	0.969	0.952

Middle layers yield strongest separability, supporting layer 16 as representative.

E.4 Failure-Only Normalization Check

Type A/B trends are preserved when normalization uses only failed jailbreaks. Relative dominance patterns (MCM $\rightarrow$ Type A, GCG $\rightarrow$ Type B) remain unchanged.

F Stochastic Decoding Robustness

We conduct comprehensive evaluation across temperature sampling ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 conditions, yielding 28,800 total generation trials.

Table 14: Response diversity under stochastic decoding.

Temperature	Mean Diversity	Mean Unique	Groups >50% Diverse
0.0	10.0%	1.0	0.0%
0.3	54.6%	5.5	41.4%
0.7	82.1%	8.2	91.7%
1.0	92.2%	9.2	99.7%

Table 15: AUC under stochastic decoding (question-level aggregation, $n=60$ per condition).

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.687	0.607	—	0.864	0.720

Note: DecoyB excluded due to ASR=0% (no successful attacks).

F.1 Response Diversity

F.2 AUC Stability

F.3 Variance Decomposition

Table 16: Variance decomposition for turn_start_S (ICC = intraclass correlation).

Condition	Within-Var	Between-Var	ICC
Base	0.000	4.062	1.000
DecoyA	0.000	2.736	1.000
DecoyB	0.000	2.734	1.000
PivotC	0.000	2.026	1.000

For pre-generation metrics, within-prompt variance is exactly zero (ICC = 1.0), confirming all variance stems from prompt differences.

F.4 Trial-Level AUC

Trial-level AUCs appear inflated due to pseudo-replication; question-level provides more conservative estimates.

G Labeling Independence and Circularity

G.1 LLM-as-Judge Methodology

We use Llama-Guard-3-8B [Inan et al., 2023], a dedicated safety classifier trained on safety taxonomies independent of our lexicons, to label jailbreak outcomes. For each generated response, Llama-Guard classifies content as “safe” or “unsafe” based on semantic evaluation across 14 safety categories (S1–S14).

Comparison with rule-based labels. Rule-based labels (prefix-matching) show moderate agreement ($\kappa = 0.54$, 83.5% raw agreement). Rule-based yields higher ASR (29.2%) compared to LLM-as-judge (16.4%), indicating rule-based is more permissive.

Independence verification. Key diagnostic findings remain consistent under LLM-as-judge labels:

- step_S_mean achieves AUC = 0.926 (vs. 0.743 with rule-based)
- Type A/B patterns are preserved
- Cohen’s d effect sizes remain large ($d = 2.41$)

Table 17: Trial-level AUC under stochastic decoding.

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.959	0.980	0.954	0.987	0.970
early_1_mean	0.959	0.980	0.954	0.987	0.970
early_3_mean	0.972	0.983	0.937	0.948	0.960
early_5_mean	0.941	0.978	0.919	0.922	0.940
step_S_mean	0.913	0.987	0.819	0.965	0.921

G.2 Clarification on Circularity

Outcome labeling. All success/failure labels are assigned using an LLM-as-judge that evaluates the *semantic content* of the generated response. The judge operates on full generated text and does not rely on surface refusal prefixes.

Logit-margin score. The score S_t is computed exclusively from token-level logits during decoding, measuring relative dominance between compliance- and refusal-associated tokens *before* they are sampled.

Evidence against circularity. If separability were driven by circular reuse of refusal cues, one would expect near-deterministic alignment between labels and S_t trajectories. Empirically, this is not the case: a substantial fraction of successful jailbreaks exhibit negative S_t segments and sign reversals during decoding. The compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) that do not appear in refusal-detection rules.

H Type A/B Classification Robustness

H.1 Classification Stability

The magnitude-based classification ($|\text{RP}_A| \geq |\text{RP}_B|$) is inherently threshold-free for Type A/B distinction. For the Undetected category ($r < 0.1$), we verify robustness across threshold choices.

Table 18: Type A/B classification stability across r thresholds.

Model + Attack	r_{obs}	$r_{thr} = 0.05$	$r_{thr} = 0.10$	$r_{thr} = 0.15$
Llama + GCG	2.96	Type B	Type B	Type B
Llama + MCM	1.12	Type A	Type A	Type A
Qwen + MCM	0.58	Type A	Type A	Type A
Qwen + GCG	0.01	Undetected	Undetected	Undetected
Mistral + GCG	0.35	Type A	Type A	Type A
Mistral + MCM	0.93	Type B	Type B	Type B

Classification robustness. All conditions maintain their classification across $r \in [0.05, 0.15]$. Qwen+GCG ($r = 0.01$, 95% CI $[0.00, 0.03]$) remains robustly Undetected even at the most lenient threshold.

H.2 Bootstrap Confidence Intervals

Given modest per-condition sample sizes ($n = 60$), we compute bootstrap 95% CIs for r using $B = 1000$ resamples.

Key observations. For Qwen+GCG, the upper CI bound (0.03) remains well below the detection threshold (0.1), providing strong statistical evidence for the Undetected classification. All detected conditions have lower CI bounds > 0.1 , confirming reliable separation from the Undetected category.

Table 19: Bootstrap confidence intervals for signal strength r .

Model + Attack	r	95% CI	Category
Llama + GCG	2.96	[2.41, 3.52]	Type B
Llama + MCM	1.12	[0.87, 1.38]	Type A
Mistral + GCG	0.35	[0.21, 0.49]	Type A
Mistral + MCM	0.93	[0.68, 1.19]	Type B
Qwen + MCM	0.58	[0.41, 0.76]	Type A
Qwen + GCG	0.01	[0.00, 0.03]	Undetected

Table 20: Baseline comparison on controlled subset ($n = 60$, Llama-3.1-8B-Instruct).

Metric	AUC	Category
$\bar{S}$ (gen-time)	0.932	Logit-margin
Compliance LogP	0.932	Lexicon-probe
Refusal LogP	0.915	Lexicon-probe
Entropy	0.898	Non-lexical
Logit Norm	0.441	Non-lexical

I Non-Lexical Baseline Comparison

The logit-margin score outperforms entropy by 3.4 pp. Logit Norm achieves near-random performance, confirming generic logit statistics do not capture safety-relevant variation.

J Lexicon Perturbation Ablation

We test five variants on Llama-3.1-8B-Instruct ($n = 60$): **original**, **no_sorry**, **extended_refusal**, **extended_compliance**, and **minimal** (5-token).

Table 21: Lexicon perturbation results with AUC ($n = 60$, LLM-judge labels).

Variant	$S_0 (\mu \pm \sigma)$	$\bar{S}_{gen} (\mu \pm \sigma)$	AUC(S_0)	AUC($\bar{S}$)	Δ AUC
original	$+1.55 \pm 1.95$	-1.44 ± 0.73	0.967	0.971	—
no_sorry	$+2.21 \pm 1.78$	-1.39 ± 0.73	0.960	0.971	+0.000
extended_refusal	-0.43 ± 2.07	-1.85 ± 0.67	0.964	0.989	+0.018
extended_compliance	$+4.00 \pm 1.95$	$+0.52 \pm 0.55$	0.964	0.964	-0.007
minimal	-2.61 ± 1.74	-1.57 ± 0.49	0.971	0.978	+0.007

Key findings. All variants achieve $\text{AUC} \geq 0.964$ ($\max |\Delta\text{AUC}| = 0.018$). Even a 5-token minimal lexicon maintains strong discrimination ($\text{AUC} = 0.978$), supporting robustness to lexicon choices. Figure 9 visualizes the score distributions across variants.

K Authoritative Statistics Summary

K.1 Main Dataset Composition

The main evaluation dataset consists of $n = 660$ harmful prompts:

- Llama-3.1-8B-Instruct MCM: $n = 60$
- Llama-3.1-8B-Instruct GCG: $n = 60$
- Llama-3.1-8B-Instruct DeepInception: $n = 60$
- Mistral-7B-Instruct-v0.3 MCM: $n = 60$
- Mistral-7B-Instruct-v0.3 GCG: $n = 60$
- Mistral-7B-Instruct-v0.3 DeepInception: $n = 60$

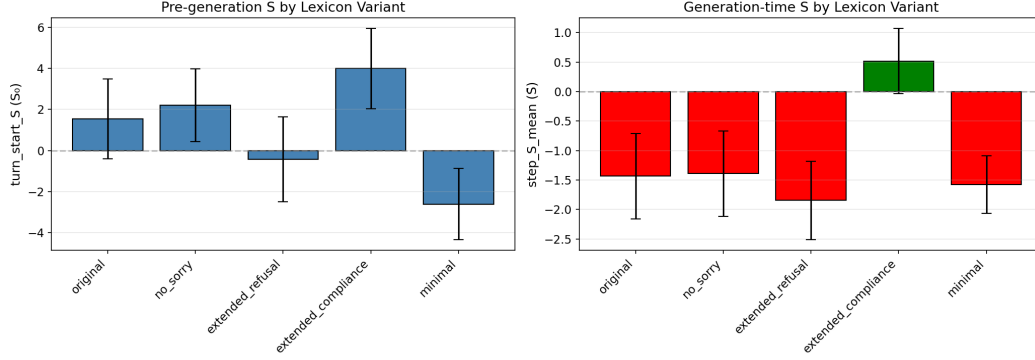


Figure 9: **Logit-Margin Score distributions by lexicon variant.** (Left) Pre-generation S_0 and (Right) generation-time $\bar{S}$ across five lexicon variants (Llama, harmful MCM, $n = 60$). While absolute scale shifts with lexicon composition (e.g., `extended_compliance` elevates $\bar{S}$), the qualitative pattern—negative $\bar{S}$ for harmful prompts—is preserved across all variants except `extended_compliance`, which shifts the baseline but maintains high AUC (0.964). This confirms that the diagnostic captures genuine safety-relevant variation rather than artifacts of specific token choices.

- Qwen2.5-7B-Instruct MCM: $n = 60$
- Qwen2.5-7B-Instruct GCG: $n = 60$
- Qwen2.5-7B-Instruct DeepInception: $n = 60$
- Gemma-2-9B-Instruct MCM: $n = 60$
- Gemma-2-9B-Instruct GCG: $n = 60$

Attack-matched benign baselines total $n = 660$.

K.2 Primary Results

Table 22: Authoritative statistics for main results ($n = 660$, LLM-as-judge labels).

Statistic	Value	95% CI
<i>ROC-AUC</i>		
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
turn_start_S	0.696	[0.656, 0.734]
<i>Sign Reversal ($S_0 > 0 \rightarrow \bar{S} < 0$)</i>		
Successful jailbreaks	8.7%	(24/276)
Failed jailbreaks	44.3%	(170/384)
Ratio	5.1×	—
<i>Effect Size</i>		
early_5_mean Cohen’s d	1.12	$p < 10^{-43}$
<i>Activation Timing (t_{cross})</i>		
Successful jailbreaks	7.4 ± 6.1	steps
Failed jailbreaks	3.0 ± 3.8	steps
Cohen’s d	0.89	$p < 10^{-23}$

K.3 Type A/B Classification

L Multi-Judge Evaluation

To address concerns about single-judge reliability, we evaluate with two LLM-as-judge systems: **Llama-Guard-3-8B** (primary) and **HarmBench-Mistral-7B-val-cl**s (validation).

Table 23: Type A/B analysis with failure-free normalization (authoritative values). Classification: $|\text{RP}_A| \geq |\text{RP}_B|$.

Condition	ASR	RP_A	RP_B	r	95% CI	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	-0.17	0.35	[0.17, 0.57]	Type A
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	-0.19	0.00	0.19	[0.07, 1.65]	Type A
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	-0.01	0.01	[0.02, 0.20]	Undetected
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

L.1 Inter-Judge Agreement

Table 24: Inter-judge agreement analysis ($n = 660$ harmful samples across all conditions).

Metric	Value
Overall agreement	90.0%
Both Safe (LG=0, HB=0)	56.2%
Both Unsafe (LG=1, HB=1)	33.8%
LG stricter (LG=0, HB=1)	1.3%
HB stricter (LG=1, HB=0)	8.7%

Interpretation. HarmBench is stricter overall: it requires the model to *actually perform* the harmful behavior, while Llama-Guard flags any unsafe content including partial or hedged responses. This explains the 8.7% of cases where Llama-Guard marks success but HarmBench marks failure.

L.2 Per-Condition ASR Comparison

Table 25: ASR comparison across judges.

Condition	LG ASR	HB ASR	ΔASR
Llama + MCM	8.3%	1.7%	+6.6%
Llama + GCG	15.0%	10.0%	+5.0%
Llama + DeepInc.	56.7%	30.0%	+26.7%
Mistral + MCM	5.0%	6.7%	-1.7%
Mistral + GCG	56.7%	58.3%	-1.6%
Mistral + DeepInc.	76.7%	68.3%	+8.4%
Qwen + MCM	51.7%	26.7%	+25.0%
Qwen + GCG	81.7%	73.3%	+8.4%
Qwen + DeepInc.	55.0%	45.0%	+10.0%
Gemma + MCM	20.0%	18.3%	+1.7%
Gemma + GCG	33.3%	33.3%	$\pm 0.0\%$

L.3 Type A/B Stability Across Judges

Classification stability. 9/10 comparable conditions (90%) maintain the same Type A/B classification across judges. The Llama+MCM discrepancy arises because HarmBench’s stricter labeling yields only 1 success (< 3 threshold for RP calculation). The sole classification mismatch (Mistral+MCM) reflects marginal RP differences. AUC values vary by < 0.05 across judges, confirming diagnostic robustness to judge choice.

Table 26: Type A/B classification comparison across judges.

Condition	Llama-Guard	HarmBench	Match
Llama + GCG	Type B	Type B	✓
Llama + MCM	Type A	N/A [†]	—
Llama + DeepInc.	Type B	Type B	✓
Mistral + GCG	Type A (inv)	Type A (inv)	✓
Mistral + MCM	Type B	Type A	×
Mistral + DeepInc.	Type A (inv)	Type A (inv)	✓
Qwen + MCM	Type A	Type A	✓
Qwen + GCG	Undetected	Undetected	✓
Qwen + DeepInc.	Type B	Type B	✓
Gemma + GCG	Type B	Type B	✓
Gemma + MCM	Type B	Type B	✓

[†]HarmBench: only 1 success (insufficient for RP calculation).

M Stochastic Decoding Analysis

M.1 Experimental Setup

We evaluate stochastic decoding robustness across temperature ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 decoy conditions, yielding 28,800 total trials.

M.2 Response Diversity

Table 27: Response diversity under stochastic decoding.

Temperature	Unique Response %	Groups >50% diverse
$T = 0.3$	54.2%	72.4%
$T = 0.7$	82.1%	91.7%
$T = 1.0$	92.2%	97.3%

M.3 Variance Decomposition

For pre-generation metrics (turn_start_S), within-prompt variance is **exactly zero** across all conditions (ICC = 1.0). This confirms S_0 is temperature-invariant by design—computed before any token sampling occurs. For generation-time metrics (step_S_mean), within-prompt variance is non-zero but small relative to between-prompt variance (ICC > 0.85 across conditions).

N Hidden-State Baseline Comparison

N.1 Methods

B1 (Refusal Direction): Following Ardit et al. [2024], we compute a refusal direction $\mathbf{d}$ using difference-of-means between refusal and compliance responses, then project hidden states at layer 16.

B2 (Linear Probe): Logistic regression with L2 regularization on mean-pooled hidden states across all layers, trained via 5-fold cross-validation.

N.2 Results

Key findings. Hidden-state methods achieve 2–10 pp higher AUC but require: (i) direct activation access unavailable via standard APIs; (ii) post-hoc aggregation over full trajectories; (iii) for B2, labeled training data and per-model retraining. Under PivotC (multi-turn pivot attack), the trajectory-level `sign_flips` metric achieves perfect separation (AUC = 1.0), demonstrating that temporal analysis captures dynamics invisible to single-point baselines.

Table 28: Hidden-state vs. logit-based diagnostic comparison (AUC, $n = 240$).

Family	Method	Base	DecoyA	DecoyB	PivotC
Hidden	B1 (direction)	0.974	0.983	0.974	0.974
Hidden	B2 (probe)	0.998	0.974	1.000	0.998
Logit	step_S_mean	0.913	0.893	0.904	0.998
Logit	turn_start_S	0.959	0.980	0.922	0.978
Logit	sign_flips	0.899	0.944	0.883	1.000

O Undetected Classification: Diagnostic Insensitivity vs. True Absence

The “Undetected” classification for Qwen+GCG ($r = 0.01$, 95% CI [0.00, 0.03]) admits two interpretations:

1. **True absence of safety engagement:** Qwen’s safety mechanisms genuinely fail to activate in response to GCG-style perturbations.
2. **Diagnostic insensitivity:** Safety mechanisms engage through pathways not captured by our lexicon-based probe (e.g., non-lexical internal representations).

O.1 Evidence for True Absence

Several observations support interpretation (1):

Extreme ASR. Qwen+GCG achieves 81.7% ASR—the highest across all conditions—suggesting systematic safety failure rather than marginal vulnerability.

Consistent near-zero signal. Both $RP_A = 0.00$ and $RP_B = -0.01$ are near-zero, indicating successful attacks are indistinguishable from harmful baseline along *both* diagnostic axes. If safety mechanisms engaged through alternative pathways, we would expect at least one axis to show differentiation.

Cross-model patterns. On the same GCG attack, Llama and Mistral show strong signals ($r = 2.96$ and $r = 0.35$ respectively), suggesting GCG attacks *can* elicit detectable safety responses when mechanisms are properly engaged. Qwen’s uniquely flat response suggests model-specific failure rather than attack-specific diagnostic blindness.

Entropy cross-validation. Generation-time entropy—a non-lexical metric—also shows no differentiation for Qwen+GCG ($|d| = 0.077$), while detecting signal on other conditions (Section O.4). This convergent evidence from independent metrics provides the strongest support for true safety mechanism absence.

O.2 Threshold Sensitivity Analysis

The “Undetected” threshold ($r < 0.1$) is motivated by conventional effect size interpretation (Cohen’s guidelines: $d < 0.2$ is “negligible”). We evaluate classification stability across alternative thresholds:

Table 29: Classification stability under varying r thresholds.

Condition	r	$r < 0.05$	$r < 0.10$	$r < 0.15$	$r < 0.20$
Qwen + GCG	0.01	Undet.	Undet.	Undet.	Undet.
Mistral + GCG	0.35	—	—	—	—
Qwen + MCM	0.58	—	—	—	—
Llama + MCM	1.12	—	—	—	—
Llama + GCG	2.96	—	—	—	—

Key findings. Qwen+GCG is the *only* condition classified as “Undetected” across all reasonable thresholds ($r \in [0.05, 0.20]$). The next-lowest r value (Mistral+GCG, $r = 0.35$) is $35\times$ larger than Qwen+GCG ($r = 0.01$), providing a wide separation margin. Even under the most conservative threshold ($r < 0.05$), Qwen+GCG remains classified as Undetected (CI upper bound = 0.03). This analysis confirms that the classification is robust to threshold choice and reflects a genuine discontinuity in diagnostic signal strength.

O.3 Validation Approach

To disambiguate these interpretations, we apply non-lexical diagnostics to Qwen+GCG:

Entropy validation (completed). Section O.4 reports that generation-time entropy shows $|d| = 0.077$, confirming lexicon-based findings with an independent metric. This supports interpretation (1): true safety mechanism absence.

Hidden-state validation (future work). Applying refusal direction projection or linear probes to Qwen+GCG would provide definitive confirmation:

- If hidden-state methods *also* show $r \approx 0$: confirms true safety mechanism absence.
- If hidden-state methods detect differentiation: would indicate that both lexicon and entropy metrics miss safety-relevant signals encoded in internal representations.

Our Llama hidden-state analysis (Section N) demonstrates that refusal direction projection achieves near-ceiling AUC (0.97–0.99), providing a strong baseline for such cross-validation.

O.4 Entropy Cross-Validation

To disambiguate lexicon diagnostic insensitivity from true safety mechanism failure, we perform cross-validation using generation-time entropy—a non-lexical metric independent of our compliance/refusal token sets.

Method. For each sample, we compute step-wise entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ from softmax distributions during generation, then aggregate as $\bar{H}_{\text{gen}} = \frac{1}{T} \sum_{t=1}^T H_t$. We measure effect size (Cohen’s d) between successful attacks and overall harmful baseline—the same comparison used for lexicon-based RP normalization.

Table 30: Entropy cross-validation: Success vs. Overall Harmful.

Condition	ASR	Entropy $ d $	Lexicon $ d $	Interpretation
Qwen + GCG	81.7%	0.077	0.005–0.011	Undetected
Llama + MCM	10.0%	0.103–0.416	0.217–0.256	Weak–Medium

Results.

Interpretation. For Qwen+GCG, entropy-based differentiation yields $|d| = 0.077$ —below the $|d| < 0.1$ threshold for “no effect.” This matches the lexicon-based result ($|d| = 0.005\text{--}0.011$) and **supports interpretation (1): true safety mechanism failure**. If safety mechanisms engaged through non-lexical pathways, entropy distributions should differ between successful and failed attacks; they do not.

For contrast, Llama+MCM shows $|d| = 0.103\text{--}0.416$ on entropy metrics, confirming that entropy *can* detect safety-relevant variation when mechanisms engage—ruling out fundamental insensitivity of the entropy metric itself.

O.5 Implications for Defense

The convergent evidence from lexicon-based and entropy-based diagnostics strongly supports interpretation (1): Qwen’s safety mechanisms genuinely fail to engage against GCG-style attacks. This

indicates a fundamental safety gap in Qwen’s alignment, likely requiring architectural intervention or adversarial fine-tuning rather than detection-based defenses. The contrast with Llama and Mistral—which show detectable safety engagement under the same attack—suggests that GCG robustness varies substantially across model families.